Sahaj Bhadja

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Summary

Software engineer with 2+ years building Python backends, agentic AI workflows, and customer-facing full-stack systems. Strong in LLM-integrated tooling, evaluation pipelines, fast iteration, and production APIs across startup and research environments.

Education

University of Southern California, Los Angeles, CA **August 2024 -- May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 -- May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 -- Present*
- Architected multi-tenant platform services on Azure SQL and Blob Storage for 500+ users, tuning indexed queries and background workflows to reduce dashboard latency from 1.2 seconds to 350 milliseconds reliably.
- Built fault-tolerant background execution for 1,000+ monthly AI jobs using explicit state transitions, retry handling, and failure isolation, improving processing throughput by 3x across concurrent production workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 -- August 2025*
- Engineered Flask services, REST APIs, and React workflows handling 10,000+ daily requests, integrating secure file ingestion and automated grading pipelines for customer-facing educational workflows in production.
- Designed multithreaded batch processing for 1,000+ uploaded files using AWS RDS-backed services, reducing grading turnaround by 60% while improving throughput, reliability, and debugging clarity during peak usage.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 -- May 2025*
- Built Python evaluation pipelines across 100,000+ trajectory records, producing reproducible metrics, logs, and visual summaries that strengthened experiment traceability and debugging for algorithmic decision workflows.
- Documented experiment behavior, failure cases, and reporting outputs for researchers, turning ad hoc testing into repeatable processes with clearer handoffs, faster iteration, and more reliable benchmarking across studies.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 -- July 2024*
- Engineered cross-platform API and ETL services using Python, SQL, AWS Glue, and S3, improving response times by 30% while supporting resilient data movement and validation across production workflows.
- Maintained Jenkins CI pipelines and 20+ automated test suites with PyTest and Mockito, increasing release reliability by 68% and improving debugging, regression control, and delivery confidence across environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 -- August 2023*
- Built REST API and WebSocket features supporting 2M+ synchronized records across modular workflows, improving data consistency and real-time responsiveness for product operations dependent on cross-service coordination.
- Automated Docker-based build and test workflows with Jenkins CI/CD, cutting release time by 50% while improving deployment consistency, troubleshooting speed, and shared validation across staging environments.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++, Bash
AI & Agentic: LLMs, prompt engineering, evaluation pipelines, OCR workflows, multimodal systems, retrieval workflows
Frontend: React, TypeScript, Next.js, responsive interfaces, customer-facing workflows
Backend & APIs: Flask, FastAPI, Node.js, REST APIs, WebSockets, microservices, background workers
Data & Cloud: PostgreSQL, MySQL, SQL Server, AWS, Azure, S3, Azure SQL, AWS Glue, Docker, Jenkins, CI/CD, Git, Agile

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript application with Docker-backed workflows.

Big-Brain Live | *LLaMA-2, FAISS, GCP, PostgreSQL*
- Built a secure document platform over 50K+ records with vector retrieval and summarization, enabling semantic search and faster document analysis through scalable cloud-backed ingestion and indexing pipelines.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a containerized RESTful data processing API on AWS with automated CI/CD releases, validation checks, and production-grade reliability for scalable backend service delivery.